

Yusuf Afifi

+447717399868 | +1(650)272-4135 | yusufafifi.com | yusuf.afifi@gmail.com | linkedin.com/in/yusuf-aff1/ | github.com/affi-yusuf | 184 Fir St San Carlos CA 94070

EDUCATION

University College London

London, UK

Bachelor of Science (Honours) in Computer Science, Minor in Applied Mathematics

September 2023 – June 2026

- First Class Honours (Expected)
- Teaching Assistant for COMP0002 in C, COMP0004 in Java, ENGF0034 in Python, and COMP0016 Systems
- Executive Quant Ventures at UCL Fintech Society

JESS Dubai

Dubai, UAE

International Baccalaureate Diploma Programme: 43/45

August 2021 – May 2023

EXPERIENCE

Software Development Engineer Intern | Gen AI, AWS, Cloud, AI Agent

London, UK

Amazon Prime Video

June 2025 – September 2025

- Reduced ad-hoc query and report generation time by **75%** by deploying a text-to-SQL agentic AI solution for Prime Video SVOD accounting team. Engineered a **RAG** workflow using **AWS Bedrock** hosting **Claude** and a **Kendra** index (containing DB schemas and business logic) to ground the agent before executing queries on **Redshift**.
- Launched a production-grade internal web application, available Amazon-wide, featuring a **React** frontend and a **Lambda** proxy handling a **WebSocket API GW** for a chatbot interface. Automated infrastructure provisioning using **CI/CD** and **AWS CDK**, while securing all contract data with least-privilege **IAM** policies.
- Curated a complete **CloudWatch** observability stack with dashboards and automated alarms on performance and cost metrics (e.g., Bedrock token usage, Lambda latency), ensuring high availability for an application supporting **200+ weekly active users** across global accounting teams.

Technology Summer Analyst | Data, ML for Finance, Climate Risk

London, UK

Angel Lane Partners (ALP Tech)

June 2024 – August 2024

- Worked on the 'GreenGuard' climate risk project to quantify asset exposure to physical, transition, and macroeconomic climate risk.
- Estimated missing data points using ML pre-processing; **KNN** and **K-means algorithms**.
- Leveraged **MATLAB** and **Python** to simulate asset value and probability of default using Merton's Model and **Vector Auto-Regression**, to model the impact of physical, transition, and macroeconomic climate risk.
- Built a static frontend to present climate stress test forecasts to clients across various climate scenarios.

PROJECTS AND RESEARCH

🌀 Petals: AI-powered Health & Wellness app | iOS Development, On-Device AI

July 2025

- Published the **first** iOS app that leverages **Apple Foundation Models** for private, on-device inference on the Neural Engine, featuring a **RAG** framework that securely accesses **Apple HealthKit** data for context.
- Engineered an instruction-tuned **LLM** chatbot ("Petal") that dynamically adapts its responses to user health and activity data, ensuring complete privacy through exclusive **on-device AI** processing.
- Powering a suite of AI-driven features—including personalized meditations, AI journaling, and curated wellness plans through advanced **context engineering** and feature integration with the foundation model.

🌀 ReadingStar: Accessibility AI Karaoke Game | On-Device Voice AI, Intel, Windows App

April 2025

- Led the development of a gamified Windows karaoke app with **Intel** & the **National Autistic Society** to aid speech development in neurodiverse students, achieving national school distribution (e.g. Helen Allison) and a feature by **Intel** as the first education app to run on an Intel NPU.
- Engineered a real-time, fully offline AI pipeline for speech-to-text, audio-to-lyric alignment, and accuracy scoring, optimized with lightweight models for low-latency inference directly within a **React Native** Windows application.
- Implemented the AI services using a **FastAPI** backend, utilizing **OpenAI Whisper** for robust offline speech recognition and the **Intel OpenVINO** toolkit to optimize model inference on target NPUs.

🌀 Machine Learning and Data Assimilation for Fluid Dynamics | Research, ML for CFD

June 2024

- Part-time undergraduate data science research project supervised by an Imperial College Research Associate.
- Constructed a **convolutional autoencoder** for accurate reconstruction of shallow water simulations using **TensorFlow**.
- Integrated **LSTM** network within latent space, to handle irregular time steps and capture temporal dependencies.
- Applied the novel ML architecture to simulate shallow water perturbations for environmental analysis.